

Generative AI: How It Works and Why It Fails

Seminar 2 • Playful Programming and Artificial Intelligence

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You already ran the model this morning

In the storytelling game, you **were** the mechanism:

- One word at a time → **next-token prediction**
- Voting on the likeliest word → **probability distribution**
- The die roll → **temperature**
- Fluent nonsense about Greenland → **hallucination**

This hour connects your game to the real thing — and shows where the game was too kind.

Part 1 • How it works

Four ideas, no equations

Idea 1 – Text becomes tokens

The model never sees words. It sees **tokens**: word fragments.

fotosyntese → fo tos yn tese

- Every token is just a number
- ~50,000–100,000 tokens cover most languages – unevenly
- Danish costs more tokens than English (worth knowing when tools behave worse in Danish)

Correction to the game: you predicted words; the model predicts fragments.

Idea 2 – Meaning becomes geometry

Each token becomes a point in a space with thousands of dimensions.

- Similar meanings sit near each other
- Directions carry meaning: the classic *king – man + woman $\approx$ queen*
- Nobody designed this space — it **emerged** from prediction practice on enormous text

The model's "understanding" is location and distance. Nothing else is in there.

Idea 3 – Attention decides what matters

For every token, the model asks: **which earlier tokens matter for predicting the next one?**

*"The trophy didn't fit in the suitcase because **it** was too big."*

To continue well, *it* must attend to *trophy*, not *suitcase*.

- This runs in parallel, across dozens of layers, for every token
- It is the reason output stays coherent across paragraphs
- **Correction to the game:** your voting used gut feeling; the model uses billions of learned attention patterns

Idea 4 – Generation is repeated sampling

The full loop, and the only loop:

1. Read all tokens so far
2. Produce a probability for **every possible next token**
3. **Sample** one (temperature = how adventurous the sample is)
4. Append it. Go to 1.

There is no plan, no fact-check, no lookup, no "mind" holding the answer.

A fluent paragraph is thousands of runs of this loop.

What training did (in one slide)

- **Pre-training:** predict the next token, across a large fraction of the written internet. This is where fluency and knowledge-shaped patterns come from.
- **Fine-tuning:** human feedback teaches it to be *helpful, harmless, polite* — this shapes the confident, agreeable tone.

Note the trade: fine-tuning rewards sounding right.

Nothing in either phase rewards being right.

Part 2 • Why it fails

Four failure modes that matter for your classroom

Failure 1 – Hallucination is the mechanism

Not a bug. Not lying. **Prediction without a world.**

- Round 4 this morning: your room produced fluent fiction about Greenland, because plausibility was the only rule
- The model is in Round 4 **all the time**
- Most dangerous where text sounds most authoritative: citations, dates, numbers, quotes

Classroom translation: fluency is not evidence. Train the reflex to check.

Failure 2 — Bias is inherited, then polished

The training data is the written internet: its perspectives, gaps, and stereotypes included.

- Under-represented in the data → poorly served by the model (languages, cultures, viewpoints)
- Fine-tuning smooths the surface — the bias returns in *which* examples, names, and defaults the model reaches for
- Danish gymnasium relevance: the model's "neutral" is anglophone

Classroom translation: ask *whose text taught it that?*

Failure 3 — Opacity, even to its makers

We can inspect every number in the network and still not explain **why this answer**.

- Billions of parameters; no human-readable rules inside (your sticky-note tree was more transparent than GPT)
- Explanations the model gives for its own answers are **also generated text** — predictions of what an explanation sounds like

Classroom translation: "ask the AI why" does not do what students think it does.

Failure 4 — Confidence is a style, not a signal

The model has no internal measure of certainty that reaches the words.

- Fine-tuning selected for the confident, helpful register
- Wrong answers arrive in the same tone as right ones
- Humans read confidence as competence — **that reflex is the vulnerability**

This afternoon's activity weaponises exactly this: *the lying, confident machine*.

The four failures are one failure

Surface	Underneath
Hallucination	prediction has no world to check against
Bias	prediction inherits its training text
Opacity	prediction is geometry, not rules
False confidence	prediction was tuned to please

One sentence for your students:

It predicts what an answer looks like.

Everything on this slide follows from that sentence.

Part 3 • So what do we teach?

Honest simplifications that survive contact

Claims you can teach that stay true at every depth:

- "It predicts plausible next words, one at a time" ✓
- "It has read enormous amounts of text, unevenly" ✓
- "It cannot check facts; it can only sound checked" ✓
- "Confidence and correctness are independent" ✓

Claims that will need unlearning later:

- "It looks things up" ✗ · "It understands you" ✗ · "It knows when it's unsure" ✗

This afternoon

You give a micro:bit's sensor data to a language model and ask it to narrate.

Then you make it fail — on purpose, with a misleading system prompt.

Bring the four failure modes with you.

You will meet all of them before 15:00.